

# Mohar

A record of who held the exam paper, and when, that nobody can quietly change afterwards — including us.

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## 1 The problem

Question papers leak before exams. It happens often enough that most people have stopped being surprised by it.

Here is the thing most people get wrong about why. Papers do not leak because someone cracked a password or broke the encryption. They leak because a person who was *allowed* to hold the paper gave it to someone else.

In the Hazaribagh case, the school principal was authorised to be in the strong room. There was no lock to break. What was missing was any record of who had the package and when. The register was written up afterwards, by hand, and it said whatever was convenient.

### SO THE REAL QUESTION IS

Not "how do we stop a leak" — nothing can promise that. It is: when a leak happens, can we prove who held the paper at every moment? And can that proof survive the people who would most like to change it?

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## 2 Our idea, in one line

Every step in the paper's journey is written down as a signed record, in a chain, where nothing can be edited or removed later — not even by us.

Three plain ideas make that work.

### **Every record is signed**

Each device — a phone, a fingerprint reader, a centre computer — has its own secret key. When it records something, it signs it with that key. Anyone can check the signature later and know which device said it. Nobody can forge a record for a device they do not have.

### **Every record is chained to the one before it**

Each record carries a fingerprint (a "hash") of the record before it. Change any old record and every record after it stops matching. It is like a numbered register where each page includes a summary of the previous page — tearing one page out is obvious immediately.

### **Nothing is ever deleted**

Records are only ever added. The database itself does not give our software permission to update or delete them. That is not a promise we made in code; it is a permission that does not exist.

#### **WHY THIS MATTERS MORE THAN IT SOUNDS**

In a normal system, if someone deletes an inconvenient step, what is left looks perfectly normal. In ours, deleting a step leaves a hole — the chain stops adding up. **A missing record is itself evidence.**

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## 3 What we actually built

### Hardware

| PART                    | WHAT IT DOES                                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| ESP32 board             | The small computer in the exam room. It reads the fingerprint reader, signs the record with its own key, and sends it. |
| R307 fingerprint reader | Reads the officials' fingers. Matching happens inside the reader itself.                                               |
| DS3231 clock            | A battery-backed clock, so the device knows the correct time even with no internet.                                    |
| Laptop                  | Runs the record system and the control room, and its camera takes the photo.                                           |
| Phone hotspot           | Connects the ESP32 and the laptop. No internet is needed.                                                              |

### Software

| PART              | WHAT IT DOES                                                                  |
|-------------------|-------------------------------------------------------------------------------|
| The ledger        | Stores the chain of signed records. Refuses anything it cannot verify.        |
| The access engine | Decides whether a package may be opened. Runs 21 separate checks.             |
| The control room  | The website. Shows every recorded act, runs the ceremony, manages the reader. |
| The watchdog      | Notifies when a device stops reporting and records that as an event.          |

### Designed but not built

We say this openly rather than letting someone find it. A **room monitor** (door sensor, presence sensor, people counter) is designed and its firmware is written, but we did not have those sensors. An **electronic seal lock** is designed but not built — and the system says so on every unlock, instead of quietly skipping that check.

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## 4 How the fingerprint part works

This is the part people ask about most, and the answer is better than they expect.

### WE NEVER STORE A FINGERPRINT

The reader keeps the fingerprint inside its own memory. When someone presses a finger, the reader answers with just two things: a **slot number** and a **match score**.

So our record says: `slot 3 matched, score 187`. That is all. No image, no fingerprint data, ever leaves the reader.

If our whole database were stolen tomorrow, no fingerprint could leak from it — because there is no fingerprint in it.

Who slot 3 is — that is kept in a separate list, like a normal staff register. It can be corrected if someone writes it down wrong. The signed records cannot be corrected, and that is the point of keeping the two apart.

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## 5 The unlock ceremony

This is the moment that matters: the sealed packet being opened. Here is what happens, in order.

- 1 The **superintendent** presses their finger on the reader.
- 2 The reader matches it and reports the slot number and score.
- 3 The ESP32 signs that as a record and sends it.
- 4 The laptop camera takes **one photo** — only now, never before.
- 5 The **observer** presses a different finger, within 2 minutes.
- 6 A second photo is taken and recorded the same way.
- 7 The system checks that two *different* fingers were used.
- 8 All 21 checks run. If every one passes, access is granted.
- 9 The paper is decrypted, printed, and the key is destroyed.

### The photo is handled carefully

The photo itself never goes into the record. Only its fingerprint — its hash — is stored. Later, anyone can hash the photo and compare it to what was recorded at the time. If they match, that is the photo taken at that moment. If someone swaps the photo, the numbers stop matching.

This keeps the record small, and it means we are not building a photo archive of officials.

#### THE CAMERA ONLY FIRES ON A FINGERPRINT MATCH

Never on a timer. Never on movement. Never because an operator asked. That is a handful of photos per exam of two officials doing one job — not hours of footage of a school corridor. Under India's data protection law, that difference is everything.

# 6 The 21 checks

Two fingerprints are not permission. They are 3 of 21 checks. The others ask:

| THE SYSTEM ASKS                           | EXAMPLE OF A REFUSAL                     |
|-------------------------------------------|------------------------------------------|
| Was a valid key presented?                | The key expired six hours ago            |
| Is this device registered to this centre? | The device belongs to a different centre |
| Is this person on the staff list?         | This finger is not registered to anyone  |
| Is the device at the centre?              | The phone is 3 km away                   |
| Is this the right time to open it?        | The exam is 18 hours away                |
| Does the seal number match?               | A different seal is on the packet        |
| Was a photo taken?                        | The camera did not work                  |

## TWO RULES WE ARE PROUD OF

**Every check runs, every time.** The system does not stop at the first failure. An attempt that fails four checks is a very different event from one that fails on a clock being slightly wrong, and an investigator needs to see all of it.

**The attempt is recorded before the answer is given.** Even a refusal is written into the chain first. There is no way to try and fail without leaving a trace.

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## 7 The key is split four ways

The paper is encrypted. But we do not give the key to one person, because then that person is the weak point.

The key is mathematically split into **four pieces**. Any **three** of them rebuild it. Fewer than three are useless — not "harder to use", genuinely useless.

| PIECE | WHO HOLDS IT                                                                 |
|-------|------------------------------------------------------------------------------|
| 1     | The exam authority                                                           |
| 2     | A public time-lock service — releases itself at the exam time and not before |
| 3     | The superintendent, released by their fingerprint                            |
| 4     | The observer, released by theirs                                             |

So no single person can open a packet early, and no single person can stop an exam by refusing. The time-lock piece is the clever part: before the exam time, that piece *does not exist yet*. It cannot be leaked, stolen or forced out of anyone, because there is nothing to take.

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## 8 What to say in the demo

Three sentences that cover it.

### WHAT IT ADDS

The record proves what was written down. The hardware proves a specific human body was physically present at the moment the sealed packet was opened — and puts a photograph of that moment into the same unchangeable record.

### WHY IT IS LIMITED ON PURPOSE

The camera fires only during the unlock ceremony, never continuously. We photograph two officials doing one job, not a school corridor. And the fingerprint never leaves the reader — we store a match result, not a biometric.

### WHAT IT DOES NOT DO

Every one of these sensors can be fooled on its own, and we say so in our own documentation. Their value is that several independent witnesses would all have to be corrupted in the same way, at the same moment — and the one thing that cannot be quietly rewritten is the record of what each of them said at the time.

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## 9 What we do not claim

Say these before anyone asks. Someone who finds a weakness you did not mention will assume there are others you are hiding.

### FINGERPRINT READERS CAN BE FOOLED

A cheap optical reader can be fooled by a lifted print. Our records prove a body was present — not that it was the right body. A better reader in real deployment, and the photo is the cross-check.

### THE DEVICE'S KEY CAN BE EXTRACTED

With the device in your hands and enough patience, its secret key can be read out of the chip. These devices are **tamper-evident, not tamper-proof**. Their value is that going silent is itself an alarm.

### TWO FINGERS IS NOT PROOF OF TWO PEOPLE

The system checks that two *different* fingerprints were used. If one person registered two of their own fingers under two names, that check passes. That is a failure of the registration process, not of the system — and the two photos would show the same face to anyone who looked.

### AND THE ONE THAT MATTERS MOST

At Hazaribagh, the principal was *allowed* to be in that room. Fingerprints and sensors catch people who should not be there. They are close to useless against someone who is supposed to be there and betrays that trust.

Which is why the hardware is not the point. **The record is the point.** It cannot stop that person — but it makes it impossible for them to pretend afterwards that they were never there.

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## 10 The hardware, in detail

Everything from here on is the technical half. The sections above are the whole idea; these are the parts, the wiring and the code.

### What is on the bench

| PART                    | WHY THIS ONE                                                                                                           |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| ESP32-WROOM-32 devkit   | Wi-Fi and enough GPIO. Dual core at 240 MHz, which matters because signing a record takes real work.                   |
| R307 fingerprint reader | Matches on its own chip and returns only a slot and a score. Talks plain serial, so it works with any microcontroller. |
| DS3231 RTC              | Temperature-compensated, battery-backed. The device must timestamp records correctly with no network at all.           |
| Laptop                  | Runs Postgres, the ledger service and the control room. Its webcam takes the ceremony photo.                           |
| Phone hotspot           | Puts the ESP32 and the laptop on one network. No internet is used at any point.                                        |

### Wiring

| FROM             | TO        | NOTE                                                    |
|------------------|-----------|---------------------------------------------------------|
| DS3231 VCC / GND | 3V3 / GND | Fit the coin cell first                                 |
| DS3231 SDA       | GPI021    | The I <sup>2</sup> C bus                                |
| DS3231 SCL       | GPI022    |                                                         |
| R307 V+ / GND    | 5V / GND  | A true R307 wants 4.2–6 V; AS608 variants want 3.3 V    |
| R307 TXD         | GPI032    | Through a 1 kΩ / 2 kΩ divider if the module runs at 5 V |
| R307 RXD         | GPI033    | Direct, no divider needed                               |

#### THE MISTAKE THAT COSTS THE MOST TIME

Serial lines must cross. The module's transmit pin goes to the pin the ESP32 *receives* on. Wire transmit to transmit and the module powers up, lights its ring, and never answers a single command — which looks exactly like a dead module.

ESP32 pins are not 5 V tolerant. If the reader runs at 5 V, its transmit line needs a divider before it reaches the board.

## Three firmware rules

- **A heartbeat every 30 seconds.** The absence of one is itself recorded as an event. Unplugging the device is not a way to go dark — it is a way to raise an alarm.
- **Write the record down before transmitting it.** Cutting the power or jamming the Wi-Fi delays a record; it does not erase one.
- **A counter that never repeats.** Each record carries a number that only goes up, kept in the chip's own storage. If the ledger holds 41 and then 44, two records existed and did not arrive — which is a different fact from two records never having been made.

#### WHERE OUR BUILD FALLS SHORT, AND SAYS SO

No SD card is fitted, so records buffer in memory and do not survive a power cut. No camera is on the board, so the photo comes from the laptop instead. Both conditions are announced into the ledger on every boot rather than quietly absorbed — a device running in a degraded state must be distinguishable in the record from one running properly.

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## 11 The software, in detail

### The pieces

| COMPONENT             | BUILT WITH                 | JOB                                                                                            |
|-----------------------|----------------------------|------------------------------------------------------------------------------------------------|
| packages/contracts    | TypeScript, Zod            | The shape of every event. One definition, shared by every service and checked at the boundary. |
| packages/crypto-core  | noble-curves, noble-hashes | Signing, hashing, the chain, the key split. Audited libraries only.                            |
| services/ledger       | Fastify, PostgreSQL        | The chain itself, plus the access engine and the registry.                                     |
| apps/control-room     | React, Vite                | The website: activity feed, ceremony page, slot register, chain verification.                  |
| firmware/witness-node | Arduino / ESP32            | Reads the finger, signs the record, serves its own control endpoint.                           |
| tools/                | Node                       | Device provisioning, the heartbeat watchdog, demo setup.                                       |

### How a record is signed

Two hashes are used, and they answer different questions.

| VALUE     | WHAT IT IS                              | WHAT IT PROVES                                                  |
|-----------|-----------------------------------------|-----------------------------------------------------------------|
| bodyHash  | Hash of the record's own contents       | Which device wrote this, and that the contents have not changed |
| chainHash | Hash of (previous chainHash + bodyHash) | That nothing has been reordered, inserted or removed since      |

The device signs only the first. It cannot sign the second, because it does not know what came before — the chain is shared, and the device may have been offline for hours. The ledger fixes the record into its position when it accepts it.

### ONE DETAIL THAT TOOK REAL CARE

Two computers must agree byte-for-byte on what was signed, or every signature fails. Ordinary JSON does not guarantee that — the same data written twice can come out with its fields in a different order, and the signature then no longer matches.

So we use a standard called RFC 8785, which fixes the field order and the number formatting. The ESP32 has no library for it, so the firmware builds its records already in that exact form and refuses to send one that came out wrong. Getting this wrong would produce a device that looks perfectly healthy and whose every record is rejected.

## What gets recorded

Twenty-one kinds of event, each with its own required fields.

| GROUP           | KINDS                                                                             |
|-----------------|-----------------------------------------------------------------------------------|
| Package journey | PACKAGE_SEALED , SEAL_APPLIED , HANDOFF , SCAN_OBSERVED                           |
| Decisions       | ACCESS_REQUESTED , ACCESS_GRANTED , ACCESS_DENIED , OVERRIDE_USED , SEAL_MISMATCH |
| The room        | MONITOR_HEARTBEAT , MONITOR_SILENT , ROOM_ENTRY                                   |
| The ceremony    | WITNESS_ASSERTED , WITNESS_CEREMONY , WITNESS_FRAME                               |
| The key         | SHARE_RELEASED , FALLBACK_INVOKED                                                 |
| Printing        | PRINT_STARTED , PRINT_COMPLETED , KEY_DESTROYED                                   |
| Anything else   | EXCEPTION_RAISED                                                                  |

## Two rules built into the event types

**Some events only a server may write.** A device cannot record that its own access was granted, and it cannot record its own silence — a device that could declare itself silent could also decline to.

**Some events need two signatures.** A handover needs a second signature from a *different* registered device. Two signatures from one device is one person pressing twice.

## The path a fingerprint takes

| STEP                    | WHERE                         |
|-------------------------|-------------------------------|
| Finger read and matched | Inside the R307               |
| Slot and score returned | R307 → ESP32 over serial      |
| Record built and signed | ESP32, using its key in flash |

| STEP                                | WHERE                                   |
|-------------------------------------|-----------------------------------------|
| Saved to the buffer                 | ESP32 memory — before any radio is used |
| Sent to the ledger                  | Wi-Fi, plain HTTP on the local network  |
| Signature verified, chained, stored | Ledger service and PostgreSQL           |
| Photo taken and its hash recorded   | Laptop browser, signed with its own key |
| Shown in the control room           | The website, updating every two seconds |

## The database enforces the important part

The application's database user has no permission to update or delete rows in the event table. The service refuses to start if it finds a connection that does. The guarantee is a database permission, not a promise made in our code — because code can be changed by whoever deploys it.

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## 12 Running it

PostgreSQL is the only thing that must be installed. No Docker, no cloud account, nothing paid.

| RUN THIS                                           | WHAT IT STARTS                    |
|----------------------------------------------------|-----------------------------------|
| <code>pnpm --filter @mohar/ledger start</code>     | The record system, on port 8081   |
| <code>pnpm --filter @mohar/control-room dev</code> | The website, on port 5173         |
| <code>tools\monitor-watchdog\run.cmd</code>        | The watchdog that notices silence |

### Three things that catch people out

- **Open the site on `localhost`** . Browsers only allow camera access on `localhost` or over HTTPS.
- **A phone hotspot changes its network numbering every restart**. The address the ESP32 was told is then wrong. It can be corrected from the website without reprogramming the device.
- **Restart the ledger after changing the event definitions**. It checks records against the version it started with.

#### WHY THE SAMPLE DATA ALWAYS REFUSES

The pre-loaded example packages cannot be granted, and that is deliberate, not broken. Their handling windows have closed, and two of the sample centres exist specifically to demonstrate a refusal. A system that granted them would be a system that had stopped checking.

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Mohar — a hackathon project. Sections 1 to 9 are the whole idea without the jargon. Sections 10 to 12 are the parts, the wiring and the code.